AN FPQC SHEAF WITH REPRESENTABLE DIAGONAL AND A SURJECTIVE FLAT UNIVERSALLY OPEN QUASI-COMPACT COVER BY A SCHEME THAT IS NOT AN ALGEBRAIC SPACE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The 'simple limit argument' for openness of Z -> T is the author's terse carried step (4389-4390); the Geraschenko mathoverflow attribution sentence (4405-4406) is kept out of the context tail, which stops at the degree-n conclusion.

Editorial note. The label lemma-not-algebraic names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 4368-4377: F(T) = the set of families (f_1, f_2, ...) of morphisms f_i: T -> P^1; P^1 satisfies the sheaf property for fpqc coverings (Descent, fpqc-universal-effective-epimorphisms) and products of sheaves are sheaves, so F is an fpqc sheaf; the diagonal is representable since T x_F S is the scheme-theoretic intersection of the closed subschemes T x_{f_i, P^1, g_i} S inside T x S
- Context span 4377-4390: SL_2 -> P^1 is surjective smooth affine; U = (SL_2)^infinity with the product morphism U -> F is an affine scheme; for f: T -> F the fibre Z = T x_F U is the infinite fibre product of the schemes Z_i = T x_{f_i, P^1} SL_2 over T; each Z_i -> T is surjective smooth affine, whence Z is a scheme flat and affine over T; a simple limit argument shows Z -> T is open, so U -> F is flat, surjective, and universally open
- Context span 4392-4405: if F were an algebraic space it would be quasi-compact and separated (by the fibre-product description), so quasi-coherent cohomology would vanish above a cutoff (Cohomology of Spaces, vanishing proposition); but pulling back O(-2, -2, ...) under the projection (P^1)^infinity -> (P^1)^n, which has a section, gives a quasi-coherent sheaf with nonzero cohomology in degree n - a contradiction

Source reference. Original lemma 023Q: sources/stacks-chow/tags.txt, source lines 2751–2751. Prerequisite pointer invoked at 4424; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 072B: sources/stacks-chow/tags.txt, source lines 9020–9020. Prerequisite pointer invoked at 4450; it names a standard fact of

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a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Consider the functor $F = (\mathbf{P}^1)^\infty$, i.e., for a scheme T the value $F(T)$ is the set of $f = (f_1, f_2, f_3, \ldots)$ where each $f_i : T \to \mathbf{P}^1$ is a morphism of schemes. Note that $\mathbf{P}^1$ satisfies the sheaf property for fpqc coverings, see Descent, Lemma 023Q. A product of sheaves is a sheaf, so F also satisfies the sheaf property for the fpqc topology. The diagonal of F is representable: if $f : T \to F$ and $g : S \to F$ are morphisms, then $T \times_F S$ is the scheme theoretic intersection of the closed subschemes $T \times_{f_i, \mathbf{P}^1, g_i} S$ inside the scheme $T \times S$. Consider the group scheme SL_2 which comes with a surjective smooth affine morphism $\mathrm{SL}_2 \to \mathbf{P}^1$. Next, consider $U = (\mathrm{SL}_2)^\infty$ with its canonical (product) morphism $U \to F$. Note that U is an affine scheme. We claim the morphism $U \to F$ is flat, surjective, and universally open. Namely, suppose $f : T \to F$ is a morphism. Then $Z = T \times_F U$ is the infinite fibre product of the schemes $Z_i = T \times_{f_i, \mathbf{P}^1} \mathrm{SL}_2$ over T. Each of the morphisms $Z_i \to T$ is surjective smooth and affine which implies that

$$Z = Z_1 \times_T Z_2 \times_T Z_3 \times_T \ldots$$

is a scheme flat and affine over Z. A simple limit argument shows that $Z \to T$ is open as well.

On the other hand, we claim that F isn't an algebraic space. Namely, if F were an algebraic space it would be a quasi-compact and separated (by our description of fibre products over F) algebraic space. Hence cohomology of quasi-coherent sheaves would vanish above a certain cutoff (see Cohomology of Spaces, Proposition 072B and remarks preceding it). But clearly by taking the pullback of $\mathcal{O}(-2, -2, \ldots, -2)$ under the projection

$$(\mathbf{P}^1)^\infty \longrightarrow (\mathbf{P}^1)^n$$

(which has a section) we can obtain a quasi-coherent sheaf whose cohomology is nonzero in degree n. Altogether we obtain an answer to a question asked

Lemma 0.1. *There exists a functor $F : Sch^{opp} \to Sets$ which satisfies the sheaf condition for the fpqc topology, has representable diagonal $\Delta : F \to F \times F$, and such that there exists a surjective, flat, universally open, quasi-compact morphism $U \to F$ where U is a scheme, but such that F is not an algebraic space.*

Proof. See discussion above. □